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ECE 6110 Lab Assignment 4
Wireless Propagation and Rate Adaptation

Part 2: Trace Rate Changes, Dropped Packets, and Goodput as User Location Varies

In this second part, we added the output of the goodput, the number of rate change and the number of packets dropped, all in each time the station is remaining at a same location. Here are the output of this simulation for a path exponent of 3.0:

Time (s)	2	4	6	8	10	12
Station Position	(10, 0, 1)	(15, 5, 1)	(20, 10, 1)	(25, 15, 1)	(30, 20, 1)	(35, 20, 1)
Distance (m)	10.44	16.09	22.56	29.31	36.18	40.42
Final Link Data Rate (Mbps)	78	78	65	58.5	39	39
Goodput (Mbps)	45.58	70.12	57.08	48.75	32.82	34.58
Rate Change	4	0	1	6	7	0
Packet Dropped	0	0	0	0	0	0

Time (s)	14	16	18	20	22	24
Station Position	(40, 20, 1)	(45, 20, 1)	(55, 20, 1)	(55, 20, 1)	(60, 20, 1)	(65, 20, 1)
Distance (m)	44.82	49.33	53.94	58.60	63.32	68.07
Final Link Data Rate (Mbps)	39	39	6.5	6.5	6.5	6.5
Goodput (Mbps)	34.43	29.67	0	0	0	0
Rate Change	0	0	14	0	0	0
Packet Dropped	0	0	7465	8649	8649	8649

As explained in part 1, Minstrel aims at setting the best possible rate in order to maximize the goodput of a transmission. Initially, as measured at time 2, the goodput is much lower than the data rate allowed by Minstrel, as the algorithm first set the lower rate available as a baseline and then tried to adapt to set a higher limit with better performance. This is why we can observe that 4 rate changes happened.

After that, the station continuously moves away from the AP, leading to a performance loss. To this end, as explained previously, it computes the success probability based on previous statistics for its fixed list of rates and select the most suitable one. This rate is a high limit, meaning that the real goodput will always be inferior, due to the channel conditions or protocol overhead. As such, as the path loss worsen, the algorithm will select a lower data rate to ensure the best transmission quality.

Moreover, around 10% of frames will be chosen for probing, meaning that they will be transmitted at a different rate in order to test channel conditions and use them to have a more accurate probability table. Furthermore, when a rate fails to deliver its packet, it will try subsequent rates. The number of times the rate was changed when the final link data is decreased comes from that. The algorithm is also conceived such that even when a rate fails, no packet are dropped by ensuring that it is very unlikely that more packets are sent than the current capacity of the system.

After some time, we observe that the goodput becomes null at the same time packets start to be dropped. This is probably due to the SNR failing below a certain threshold needed to maintain a stable connection, when the AP and the station are too far from each other. This leading to all packets being dropped. We can observe that initially less packet are dropped while Minstrel try to set up the correct data rate, at time 18, than later, as less packets are transmitted during the probing.